



Effect of Mo and Ni elements on microstructure evolution and mechanical properties of the $\text{CoFeNi}_x\text{VMo}_y$ high entropy alloys



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ARTICLE INFO

Article history:

Received 29 April 2015

Received in revised form

16 July 2015

Accepted 20 July 2015

Available online 21 July 2015

Keywords:

High entropy alloys

Crystal structure

Microstructure

Mechanical properties

ABSTRACT

A series of CoFeNiVMo_y ($y = 0-1$) and $\text{CoFeNi}_x\text{VMo}$ ($x = 1-2$) high entropy alloys were designed in order to study the effect of Mo and Ni elements on their microstructure evolution and mechanical properties. Results show that for the quinary CoFeNiVMo alloy, by either decreasing the Mo content to 0.6 or increasing the Ni content to 1.4, eutectic microstructures can be obtained. For the CoFeNiVMo_y alloys, with increasing Mo content, the volume fraction of the CoMo_2Ni -type intermetallic phase increases which results in a decrease of the plastic strain and an increase of the yield strength, and the Vickers hardness shows an approximately linear increase from HV 238.1 to HV 624.6. While for the $\text{CoFeNi}_x\text{VMo}$ alloys, with increasing Ni content, the increased FCC solid solution phase leads to an increase of the plastic strain and a decrease of the yield strength. Among these alloys, the $\text{CoFeNiVMo}_{0.2}$ alloy shows the best comprehensive compressive properties: high yield strength of 686.9 MPa and ductility above 80%. In addition, the formation condition of eutectic HEAs was also investigated. It is found that the parameters of valence electron concentration (VEC), atomic size difference (δ), enthalpy of mixing (ΔH_{mix}), entropy of mixing (ΔS_{mix}) and electronegativity ($\Delta\chi$) cannot effectively design the composition of eutectic HEAs.

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1. Introduction

Recently, a new class of alloys called high entropy alloys (HEAs) or multi-component alloys have attracted extensive research attention because of their unique physical, chemical and mechanical properties [1–11]. HEAs are generally composed of at least five principal elements, each at concentrations of 5–35 at.% [10,11]. Recent research has reported that the majority of HEAs contain multiple phases rather than a definitive single-phase solid solution, which may involve a secondary solid solution phase and/or intermetallic compounds [12–16]. The diversity of the elements in HEAs can lead to distinctive microstructures. For example, eutectic microstructures have been found in $\text{AlCoCrFeNi}_{2.1}$ [5], $\text{AlCrFeNiMo}_{0.2}$ [17], $\text{Al}_{1.2}\text{CrCuFeNi}_2$ [18] and $\text{CoFeNi}_2\text{V}_{0.5}\text{Nb}_{0.75}$ [19] HEAs. In fact, alloys with central solid solution structures often show high melting points and inferior castability. However, alloys with eutectic microstructures possess low

melting points, good castability and regular microstructures. This means eutectic alloys have great potential for industrial applications. In present, the eutectic alloys have been used to cast auto parts with complex shape, such as flange and gear. Therefore, this study aimed to design HEAs with eutectic compositions. As most reported eutectic HEAs contain Co, Cr, Fe, Ni, Mo, V and Nb elements, a novel HEAs system $\text{CoFeNi}_x\text{VMo}_y$ was designed and the effect of Mo and Ni elements on the microstructure evolution were investigated. All the elements in this alloy system have high melting points, and therefore have the potential to be applied in the aerospace industry.

2. Experimental

Button ingots of $\text{CoFeNi}_x\text{VMo}_y$ alloys (when $x = 1$, $y = 0, 0.2, 0.4, 0.6, 0.8, 1$ and when $y = 1$, $x = 1, 1.2, 1.4, 1.6, 1.8, 2$) were prepared by vacuum arc melting in Ti-gettered high-purity argon atmosphere with constituent element purity higher than 99.95 wt.%. These alloys can be denoted as Mo_y or Ni_x for short, e.g. when $y = 1$, $x = 1.2$, $\text{CoFeNi}_{1.2}\text{VMo}$ was denoted as $\text{Ni}_{1.2}$ in this paper. Subscripts in the alloy compositions (x and y) indicate the atomic portion of each individual element. To achieve a homogeneous distribution of

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elements in the alloys, each alloy was remelted and flipped five times. The prepared button-shaped specimens were approximately 30 mm in diameter and 15 mm in thickness, and had shiny surfaces, indicating that there was no oxidation during the vacuum arc melting.

X-ray diffraction (XRD) patterns were obtained using an EMPYREAN diffractometer (made in Netherlands, 2011) operated at 30 kV and 20 mA. The range of scanning angle (2θ) was from 20° to 100° and the scanning speed was $0.215629\text{ deg}\cdot\text{s}^{-1}$. A Zeiss supra 55 field-emission-gun scanning electron microscope (SEM) equipped with the energy dispersive spectrometry (EDS) was used to analyze the microstructure and chemical compositions. The compressive properties were evaluated by uniaxial compression tester (DNS-100, China, 2009) based on the ASTM E9-89a standard on samples with diameter of 5 mm and height of 10 mm at a train rate of $1 \times 10^{-3}\text{ s}^{-1}$. Multiple compressive tests were conducted for each composition according to the compressive results. Vickers microhardness was measured on polished cross-section surfaces using a 136° Vickers diamond pyramid under a 1 kg load applied for 15 s. The reported hardness values were the average of at least 6 measurements.

3. Results and discussion

3.1. Crystal structure

Fig. 1a shows the XRD patterns of the Moy alloys. As can be seen, the Mo0 alloy presents a single face-centered-cubic (FCC) crystal structure. The Mo02, Mo04, Mo06, Mo08 and Mo1 alloys show FCC phase accompanied with CoMo_2Ni -type intermetallic phase. And with increasing Mo content, the intensity of the FCC phase peaks decreases and that of the CoMo_2Ni -type intermetallic phase increases. Fig. 1b presents the XRD patterns of the Nix alloys, in which the diffraction peaks can be identified as FCC phase as well as Co_2Mo_3 -type intermetallic phase. And with increasing Ni content, the intensity of the FCC phase peaks increases and that of the Co_2Mo_3 -type intermetallic phase decreases. A conclusion can be drawn that Ni element is an FCC stabilizer, whereas Mo element promotes the formation of intermetallic phases. In fact, the near-zero mixing enthalpy of Ni element with other elements can effectively favor the formation of solid solution structures and the negative mixing enthalpy of Mo element with other elements will promote the formation of some intermetallic phases (see Table 1) [20].

Table 1
The mixing enthalpy ΔH_{mix} (kJ/mol) of atom pairs in the $\text{CoFeNi}_x\text{VMo}_y$ alloys.

	Co	Fe	Ni	V	Mo
Co	0	−1	0	−14	−5
Fe	−	0	−2	−7	−2
Ni	−	−	0	−18	−7
V	−	−	−	0	0
Mo	−	−	−	−	0

3.2. Microstructures

Typical microstructures of the Moy alloys are shown in Fig. 2. The CoFeNiV alloy presents a single solid solution structure with coarse grains of hundreds of microns, as shown in Fig. 2a. When $y = 0.2$, the microstructure is composed of a main FCC solid solution phase (marked as α) and almost continuous network of second phase at the grain boundaries (marked as β_1). With further increased Mo content, the microstructures change from a hypoeutectic microstructure (primary phase α + eutectic regions E1) for Mo04, to a fully eutectic microstructure (E1) for Mo06, and finally to hypereutectic microstructures (primary phase β_1 + eutectic regions E1) for Mo08 and Mo1 alloys. Fig. 3 shows the microstructures of the Nix alloys, as can be seen, the microstructures change from hypereutectic microstructures (β_2 + E2) for Ni1 and Ni1.2, to a fully eutectic microstructure (E2) for Ni1.4, and finally to hypoeutectic microstructures (α + E2) for Ni1.6, Ni1.8 and Ni2 alloys. It is clear that the effect of Mo and Ni elements on the eutectic microstructure evolution of the $\text{CoFeNi}_x\text{VMo}_y$ alloys shows an inverse tendency. For the Ni1/Mo1 alloy which is common owned in Moy and Nix alloys, fully eutectic microstructures can be obtained by decreasing the Mo content or increasing the Ni content. And the eutectic microstructure with large-size eutectic cells (hundreds of microns) of the Mo06 and Ni1.4 alloys are shown in Fig. 4.

To further analyze the effect of Mo and Ni elements on the eutectic microstructures, the chemical compositions of different regions of the alloys were measured by EDS. The results are presented in Table 2, with the average mole fraction of Mo element in each alloy listed on the right side. It can be found that the concentration of Mo element in the Moy alloys is generally less than that of the Nix alloys. And for both alloy systems, the dark α phase is enriched with Ni element while the light β_1 and β_2 phases are enriched with Mo element. The distribution of V, Fe and Co elements is uniform in α , β_1 and β_2 regions. The dark α regions in both alloy systems show similar chemical compositions, and combined

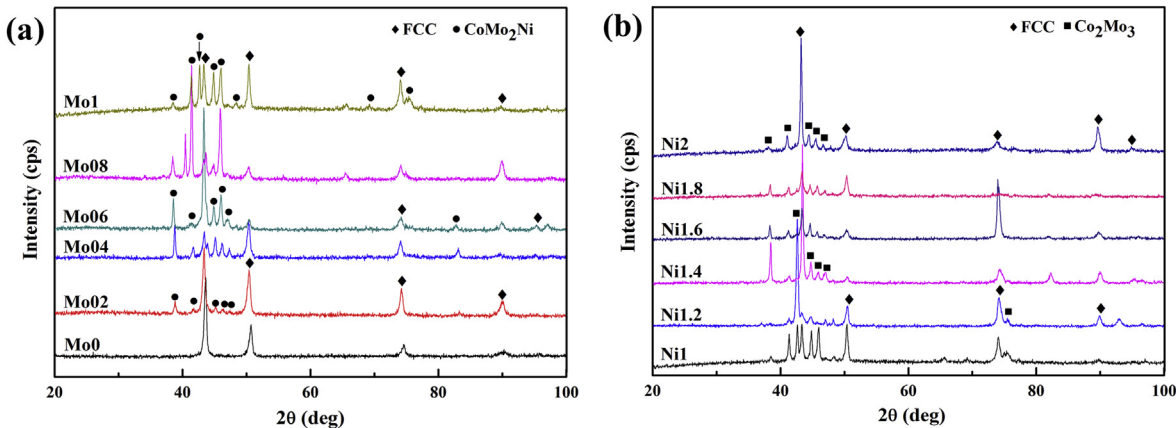


Fig. 1. XRD patterns of the $\text{CoFeNi}_x\text{VMo}_y$ alloys: (a) CoFeNiVMo_y ($y = 0; 0.2; 0.4; 0.6; 0.8; 1$), (b) $\text{CoFeNi}_x\text{VMo}$ ($x = 1; 1.2; 1.4; 1.6; 1.8; 2$).

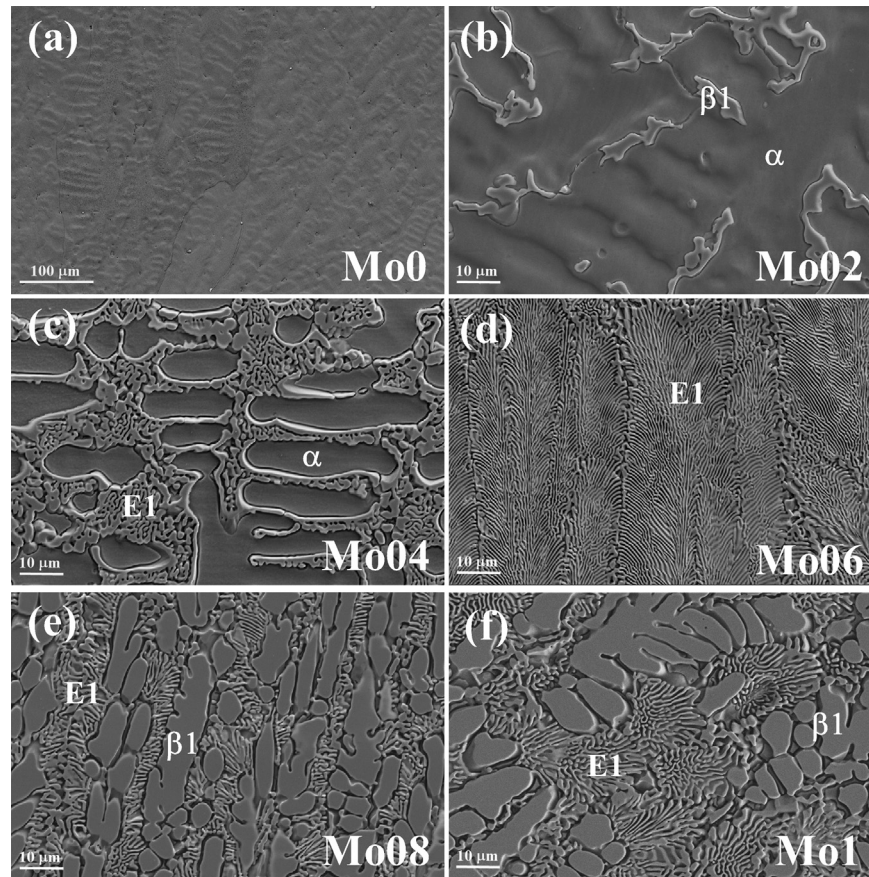


Fig. 2. SEM/SE micrographs of the CoFeNiVMoy ($y = 0; 0.2; 0.4; 0.6; 0.8; 1$) alloys.

with the XRD results, a conclusion can be drawn that the primary phase α is the FCC solid solution phase. The Mo content of the β_2 phase in the Ni1.2 alloy is obviously higher than that of the β_1 phase in the Ni1/Mo1 alloy, although the average Mo concentration in the Ni1/Mo1 alloy is higher than that in the Ni1.2 alloy. Furthermore, the Mo content of the E2 regions is obviously increased compared to that of the E1 regions. These results indicate that the intermetallic phase of the eutectic lamellar microstructures in the Moy and Nix alloys are different, and the β_2 phase in the Nix alloys are more enriched with Mo element than that of β_1 phase in the Moy alloys. Table 2 shows that the average Mo concentration of both the Mo08 and Ni2 alloys is 16.67 at.%; however, the Mo08 alloy shows a hypereutectic microstructure while the Ni2 alloy shows a hypoeutectic microstructure (see Figs. 2e and 3f). The eutectic regions of the Mo08 and Ni2 alloys own distinctively different Mo contents, i.e., 22.60 at.% in the Mo08 alloy and 35.68 at.% in the Ni2 alloy, while the width of the eutectic lamellar structure of both the Mo08 and Ni2 alloys are similar. This also indicates that the intermetallic phase composition of the eutectic lamellar microstructure in the Moy and Nix alloys are different. This well corresponds to the XRD results which show that the intermetallic phase of the eutectic lamellar microstructures in the Moy alloys is the CoMo₂Ni phase and that in the Nix alloys is the Mo-rich Co₂Mo₃ phase, as marked as β_1 and β_2 in Figs. 2 and 3, respectively. As the average mole fraction of Mo element in the Nix alloys is higher than that in the Moy alloys, the increased Mo atoms separate from the FCC solid solutions more distinctly and form the Co₂Mo₃ intermetallic phase. Moreover, in the Moy alloys, the Mo content of the E1 regions increases from 13.11 to 22.11 at.% with increasing Mo content, which promotes more Mo atoms solid solute into the E1 regions. As for the

Nix alloys, the Mo content of the E2 regions shows the same increasing tendency from 23.26 to 35.68 at.% with increasing Ni content. This is because for the hypereutectic microstructure of the Ni1.2 alloy, the Mo-rich primary phase can dissolve large amounts of Mo atoms. While for the hypoeutectic microstructures of the Ni1.4 to Ni2 alloys, the FCC primary phase will reject Mo atoms into the E2 regions which contribute to the increase of Mo content.

3.3. Mechanical properties

The microhardness curves of the Moy and Nix alloys are given in Fig. 5, and the specific values are also listed. It can be seen that the hardness of the Moy alloys increases with the addition of Mo element. The maximum value of HV 624.6 was obtained in the equiatomic CoFeNiVMo alloy. For the Nix alloys, the hardness decreases with an increased Ni content. The high hardness of the Nix alloys is attributed to the solid solution strengthening effect of larger amounts of Mo element compared to the Moy alloys. However, the hardness of the Mo08 alloy is obviously higher than that of the Ni2 alloy, although they have the same concentration of Mo element. This is because the Mo08 alloy shows a hypereutectic microstructure while the Ni2 alloy shows a hypoeutectic microstructure. And the microstructure of the Mo08 alloy is constituted by a hard β_1 primary phase (intermetallic phase) with a eutectic lamellar structure. Conversely, the microstructure of the Ni2 alloy is constituted by a soft α primary phase (FCC solid solution phase) with a eutectic lamellar structure. Therefore, the hardness of the Mo08 alloy is obviously higher than that of the Ni2 alloy even though they have the same Mo content.

Mechanical properties of the alloys were further studied using

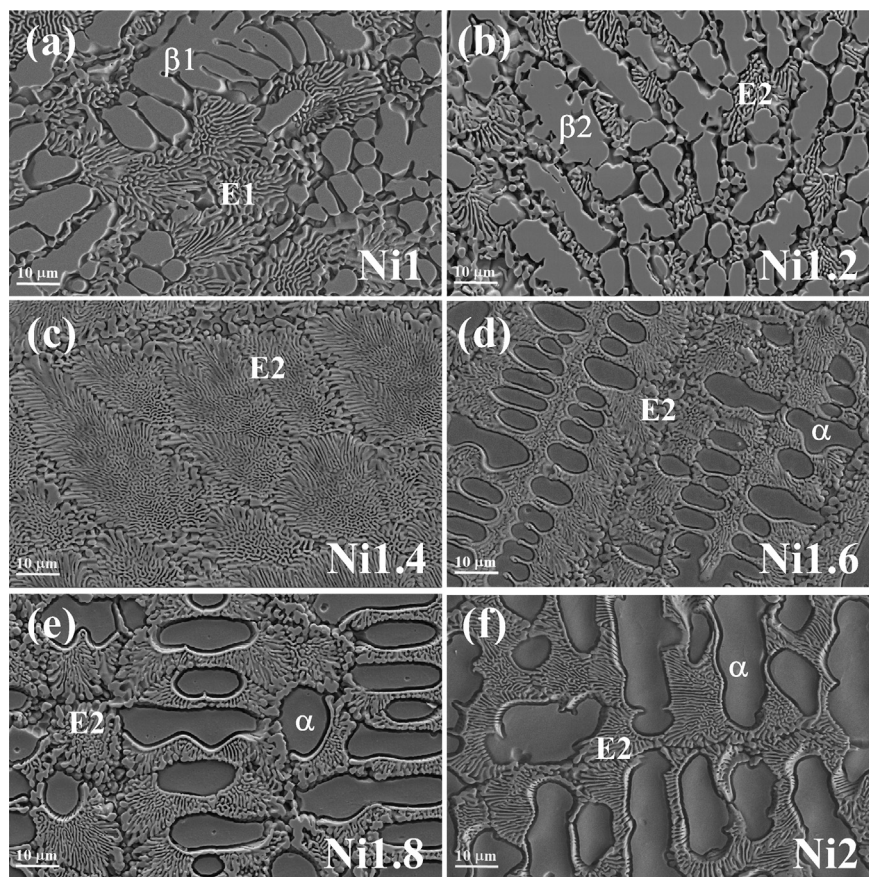


Fig. 3. SEM/SE micrographs of the CoFeNi_xVMo ($x = 1; 1.2; 1.4; 1.6; 1.8; 2$) alloys.

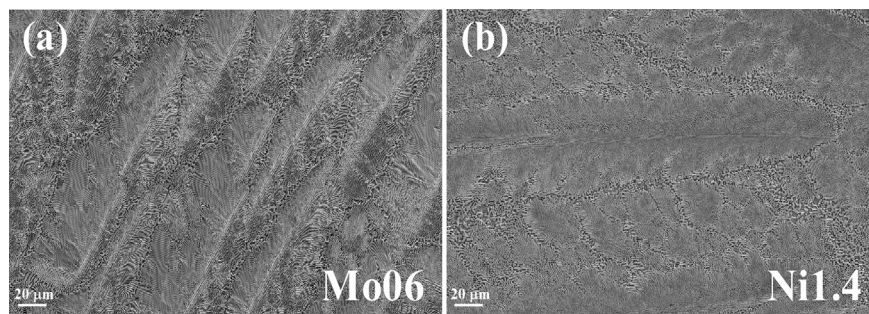


Fig. 4. SEM/SE images of the eutectic (a) CoFeNiVMo_{0.6} and (b) CoFeNi_{1.4}VMo alloys in low magnification.

compressive tests and the engineering stress–strain curves of the Mo_y alloys are shown in Fig. 6a. Evidentially, the Mo content has a very pronounced effect on the compressive behavior of the Mo_y alloys. The Mo-free alloy has a low yield strength of 428.3 MPa, and shows excellent ductility which can be compressed to 80% height reduction without fracture. The Mo02 alloy exhibits a similar behavior, although its yield strength of 686.9 MPa is slightly higher than that of the Mo-free alloy. A further increased Mo content results in the deterioration of ductility, which is evidenced by the compressive ductility of the Mo04 alloy being decreased to 19.7%. The Mo06, Mo08 and Mo1 alloys have almost no plastic ductility ($\epsilon < 5\%$) while the Mo08 alloy displays a maximum fracture strength of 2163.8 MPa. Fig. 6b illustrates the engineering stress–strain curves of the Ni_x alloys. The Ni1 to Ni1.8 alloys show similar compressive properties, with low strength and ductility. However, the Ni2 alloy shows a maximum compressive ductility of

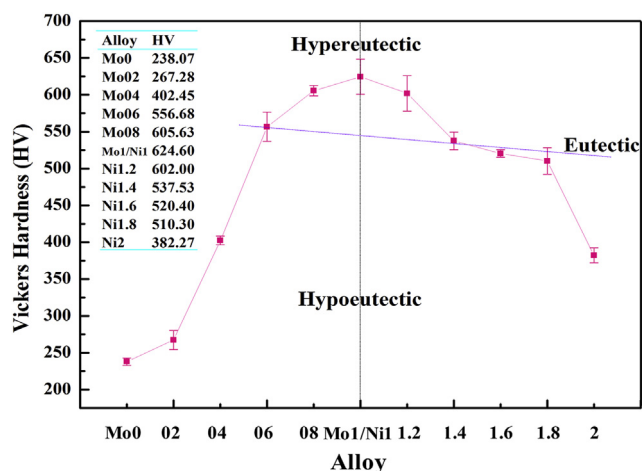
24.7% and its compressive yield strength and fracture strength are 679.1 MPa and 1152.6 MPa respectively. It is obvious that the eutectic CoFeNiVMo_{0.6} and CoFeNi_{1.4}VMo HEAs do not demonstrate the best mechanical properties, which is unlike the AlCoCrFeNi_{2.1} [5] and AlCrFeNiMo_{0.2} [17] eutectic HEAs reported previously. This is attributed to the large-size eutectic cells with straight boundaries of the Mo06 and Ni1.4 HEAs (hundreds of microns, as shown in Fig. 4). And for the small button ingot, the compressive samples only contain several eutectic cells. During compressive test, the boundaries of the eutectic cells ruptured firstly which lead to the weak plasticity and strength of the alloys.

3.4. Prediction and design of eutectic HEAs

Considering the excellent castability, regular microstructures and good high-temperature creep resistance of eutectic alloys, they are

Table 2Chemical compositions of different regions in the CoFeNi_xVMo_y alloys measured by EDS (at.%), and the average mole fraction (at.%) of Mo element in each alloy.

Alloys	Regions	Chemical compositions					Mole fraction of Mo element
		V	Fe	Co	Ni	Mo	
Mo0	α	24.98	25.26	24.22	25.54		0.00
Mo02	α	24.12	24.79	23.22	22.69	5.18	4.76
	E1	29.31	21.10	20.25	16.23	13.11	
Mo04	α	21.64	23.88	22.60	24.63	7.25	9.09
	E1	26.99	19.92	20.21	15.36	17.52	
Mo06	E1	24.31	21.24	19.64	17.49	17.33	13.04
Mo08	β 1	21.37	19.62	19.72	15.98	23.33	16.67
Mo1/Ni1	E1	22.46	19.65	18.91	16.38	22.60	20.00
	β 1	22.30	20.41	18.84	15.08	23.37	
Ni1.2	E1	23.19	19.89	19.92	14.90	22.11	19.23
	β 2	19.48	18.62	15.80	16.98	29.12	
Ni1.4	E2	20.13	19.65	19.37	17.59	23.26	18.52
	E2	20.55	17.02	17.91	16.64	27.88	
Ni1.6	α	17.12	18.56	16.91	34.59	12.83	17.86
	E2	17.70	14.91	14.82	18.62	33.96	
Ni1.8	α	17.15	18.91	18.08	33.32	12.54	17.24
	E2	16.97	15.04	15.76	18.5	33.74	
Ni2	α	15.96	17.88	17.69	35.34	13.14	16.67
	E2	16.24	14.36	15.43	18.29	35.68	

**Fig. 5.** Effects of Mo and Ni content on the Vickers hardness of the CoFeNi_xVMo_y alloys.

very promising in industrial applications. However, until now, fully eutectic composition was only found in a small number of high entropy alloys. The known eutectic HEAs are as follows: AlCoCrFeNi_{2,1}

[5], AlCrFeNiMo_{0.2} [17], Al_{1.2}CrCuFeNi₂ [18] and CoFeNi₂V_{0.5}Nb_{0.75} [19]. Thus, how to find eutectic points (compositions) in multi-component high entropy alloys becomes a significant problem. At present, several parameters based on the Hume-Rothery theory are widely used to predict the phase formation of HEAs, i.e., valence electron concentration (VEC) [21], atomic size difference (δ), mixing enthalpy (ΔH_{mix}), mixing entropy (ΔS_{mix}) [22,23] and electronegativity ($\Delta\chi$) [24]. The valence electron concentration $VEC = \sum c_i VEC_i$, in which c_i and VEC_i is the atomic percentage and valence electron concentration of the i th element, respectively. The atomic size difference $\delta = 100\% \sqrt{\sum c_i (1 - r_i/\bar{r})^2}$, $\bar{r} = \sum c_i r_i$, where r_i is the atomic radius of the i th component. The mixing enthalpy $\Delta H_{mix} = \sum_{i=1, i \neq j}^n \Omega_{ij} c_i c_j$, $\Omega_{ij} = 4\Delta_{mix}^{AB}$, where Δ_{mix}^{AB} is the mixing enthalpy of binary liquid AB alloys. The mixing entropy is $\Delta S_{mix} = -R \sum c_i \ln c_i$. The electronegativity $\Delta\chi = \sqrt{\sum c_i (\chi_i - \bar{\chi})^2}$, $\bar{\chi} = \sum c_i \chi_i$, where χ_i is the Pauling electronegativity of the i th element. The values of these parameters of the present Mo_y and Ni_x alloys as well as the four reported eutectic HEAs were calculated and summarized, as listed in Table 3. For the Mo_y and Ni_x alloys, the parameter values for the hypoeutectic, eutectic and hypereutectic alloys are very close, and no regularities can be found in the Mo_y and Ni_x alloys. The case is similar in the other four

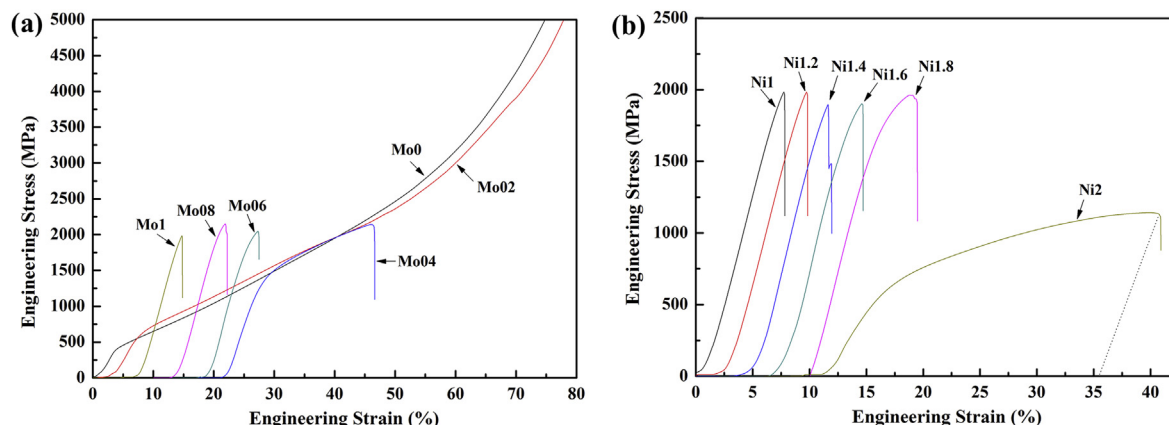
**Fig. 6.** Compressive engineering stress–strain curves of the CoFeNi_xVMo_y alloys: (a) CoFeNiVMo_y ($y = 0; 0.2; 0.4; 0.6; 0.8; 1$), (b) CoFeNi_xVMo_x ($x = 1; 1.2; 1.4; 1.6; 1.8; 2$).

Table 3

The parameters VEC , δ , ΔH_{mix} , ΔS_{mix} , and $\Delta \chi$ for the present Mo y and Ni x alloys as well as the four reported eutectic HEAs (the eutectic alloys are marked with *).

Alloys	VEC	δ	ΔH_{mix}	ΔS_{mix}	$\Delta \chi$
Mo0	8.00	3.11	−10.50	11.53	0.1092
Mo02	7.90	3.60	−10.16	11.36	0.1297
Mo04	7.82	3.95	−9.83	11.20	0.1443
Mo06*	7.74	4.20	−9.53	11.03	0.1551
Mo08	7.67	4.40	−9.24	10.87	0.1634
Mo1/Ni1	7.60	4.54	−8.96	10.70	0.1699
Ni1.2	7.69	4.54	−9.08	10.54	0.1667
Ni1.4*	7.78	4.53	−9.16	10.39	0.1636
Ni1.6	7.86	4.52	−9.21	10.23	0.1607
Ni1.8	7.93	4.50	−9.23	10.08	0.1580
Ni2	8.00	4.48	−9.22	9.93	0.1554
AlCoCrFeNi _{2.1} * [5]	7.71	5.20	−11.94	9.86	0.1204
AlCrFeNiMo _{0.2} * [17]	6.71	5.78	−12.65	11.36	0.1474
Al _{1.2} CrCuFeNi ₂ * [18]	7.84	5.40	−6.78	9.98	0.1201
CoFeNi ₂ V _{0.5} Nb _{0.75} * [19]	8.24	5.91	−15.90	10.17	0.1189

reported eutectic alloy systems (see Table 3). Recently, a feasible method (other than the experimental method), i.e., the CALPHAD approach [25], is proposed to investigate phase stability and phase relationships in a multi-component system. While development of a complete database with 10 or more elements for HEA design is extremely challenging, a two-step approach is proposed in a previous study to face this challenge. In the first step, binary phase diagrams are calculated to identify matching elements, and in the second step, a thermodynamic data base can be developed for the limited number of matching elements (details can be obtained in Ref. [25]). The calculation of phase diagram is promising in predicting the formation of eutectic HEAs and worth further investigation.

4. Conclusions

CoFeNi $_x$ VMo $_y$ (when $x = 1$, $y = 0, 0.2, 0.4, 0.6, 0.8, 1$ and when $y = 1$, $x = 1, 1.2, 1.4, 1.6, 1.8, 2$) HEAs were produced by vacuum arc melting and the effect of Mo and Ni elements on the eutectic microstructure evolution and mechanical properties were investigated. Afterwards, the following conclusions were made:

- (1) When $x = 1$, for the CoFeNiVMo $_y$ alloy system, the microstructures transform from a single FCC structure for Mo0, to a dendritic structure for Mo02, to a hypoeutectic microstructure for Mo04, then to a fully eutectic microstructure for Mo06, and finally to hypereutectic microstructures for Mo08 and Mo1 alloys. When $y = 1$, for the CoFeNi $_x$ VMo alloy system, the microstructures transform from hypereutectic microstructures for Ni1 and Ni1.2 to a fully eutectic microstructure for Ni1.4, and finally to hypoeutectic microstructures for Ni1.6, Ni1.8 and Ni2 alloys.
- (2) The eutectic CoFeNiVMo_{0.6} alloy consists of a FCC phase plus a CoMo₂Ni-type intermetallic phase, while the eutectic CoFeNi_{1.4}VMo alloy consists of a FCC phase plus a Co₂Mo₃-type intermetallic phase.
- (3) For the CoFeNiVMo $_y$ alloys, with increasing Mo content, the increased level of the CoMo₂Ni-type intermetallic phase

leads to a decrease of the plastic strain and an increase of the yield strength, and the Vickers hardness shows an approximately linear increase from HV 238.1 to HV 624.6. While for the CoFeNi $_x$ VMo alloys, with increasing Ni content, the increased FCC solid solution phase leads to an increase of the plastic strain and a decrease of the yield strength, and the Vickers hardness decreases from HV 624.6 to HV 382.3.

- (4) The present parameters for predicting HEAs, i.e., VEC , δ , ΔH_{mix} , ΔS_{mix} and $\Delta \chi$ are summarized to explore the formation condition of eutectic HEAs, but no regularities can be found. In addition to the experimental method, the CALPHAD method has potential in designing the composition of eutectic HEAs.

Acknowledgments

This work was supported by the National Natural Science Foundation of China Nos. 51134013, 51104029, 51274054, 51375070, and 51471044, the fundamental research funds for the central universities, key laboratory of basic research projects of Liaoning Province Department of Education (LZ2014007) and the Natural Science Foundation of Liaoning Province (2014028013).

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